

LECTURE 04 · REVIEW NOTES

FX Markets & Arbitrage

Exchange rate drivers, cross-rate mechanics, triangular arbitrage, and the anatomy of the 2007–2013 FX cartel.

About these notes

These notes accompany Lecture 04 of FINA 405 · International Financial Management (University of San Diego, Knauss School of Business, Fall 2026). They present the core analytical framework in textbook style. Numerical examples are verified independently; all computations appear in full. In-class activities are described by topic only — solutions are discussed during the session.

1. What Moves Exchange Rates?

The exchange rate between two currencies is determined by the supply of and demand for each currency in the global foreign-exchange market. Six forces routinely drive these flows:

Driver	Direction of Effect	Mechanism
Interest Rate Differentials	Higher-rate currency appreciates	Capital flows toward higher yields; a Fed rate surprise that lifts U.S. rates draws global investors into dollar-denominated assets.
Trade Flows	Exporter's currency appreciates	A persistent current-account surplus creates net inflows of foreign currency, bidding up the domestic currency.
Carry Trade	High-yield target appreciates; low-yield funding depreciates	Investors borrow in low-rate currencies (JPY, CHF) and invest in high-rate ones (MXN, AUD). Carry unwinds violently in risk-off episodes.
Safe-Haven Demand	USD / JPY / CHF appreciate in crises	During stress, capital flees to perceived safety. The U.S. dollar strengthened in 2008 even as the crisis originated in the United States.
Fiscal & Political Risk	Local currency depreciates	High fiscal deficits or political instability raise sovereign-risk premia. Argentina, Turkey, and Brazil have experienced repeated episodes.
Central Bank Intervention	Direction set by CB intent	Direct purchases or sales of foreign reserves. The Swiss National Bank defended a 1.20 CHF/EUR floor; the Bank of Japan intervened to slow yen depreciation in 2022.

These drivers often interact. For example, a central bank rate hike simultaneously affects interest differentials and can trigger carry-trade realignments. Understanding which force dominates a given episode is central to FX analysis.

2. Computing Cross Rates

A cross rate is the exchange rate between two currencies derived from their individual quotes against a common third currency (typically the U.S. dollar). Two arithmetic methods apply depending on how the quotes are expressed.

Method A: Division (Shared Quote Currency)

When both rates are quoted as units of the same currency per one unit of the respective base currency — e.g., both quoted in USD — the cross rate is obtained by dividing one rate by the other. The shared quote currency cancels algebraically:

$$\text{EUR/GBP} = (\text{EUR/USD}) \div (\text{GBP/USD})$$

WORKED EXAMPLE 1

EUR/GBP Cross Rate and Triangular Arbitrage Check

Input	Value	Source
EUR/USD	1.1200	Market quote
GBP/USD	1.2700	Market quote
Implied EUR/GBP	$1.1200 \div 1.2700 = 0.8819$	No-arb calculation
Market EUR/GBP	0.8740	Broker quote
Mispricing gap	$0.8819 - 0.8740 = 79 \text{ pips}$	Arbitrage signal

A 79-pip gap is substantial relative to typical EUR/GBP bid-ask spreads of 1–2.5 pips, indicating a clear triangular arbitrage opportunity.

Method B: Multiplication (Bridge Currency)

When two quotes share a common base currency that serves as a bridge, the cross rate is the product of the two rates:

$$\text{GBP/JPY} = (\text{GBP/USD}) \times (\text{USD/JPY})$$

WORKED EXAMPLE 2

GBP/JPY Cross Rate

Input	Value	Source
GBP/USD	1.2700	Market quote
USD/JPY	149.50	Market quote
Implied GBP/JPY	$1.2700 \times 149.50 = 189.87$	No-arb calculation
Market GBP/JPY	188.20	Broker quote
Mispricing gap	$189.87 - 188.20 = 167 \text{ pips}$	Arbitrage signal

The 167-pip deviation far exceeds typical GBP/JPY spreads of 2–5 pips.

3. The Mispricing Ratio

The mispricing ratio provides a quick diagnostic for arbitrage opportunities. It is defined as:

Mispricing Ratio = Implied (No-Arb) Rate $\div$ Market Quote

Ratio	Interpretation	Action
> 1	Market rate is too cheap — the implied rate exceeds the market quote	Arbitrage exists: trade the implied path against the market.
= 1	Market is fairly priced — no deviation from no-arb condition	No trade.
< 1	Market rate is too expensive	Reverse the trade direction.

Applied to Example 1: Implied EUR/GBP = 0.8819; Market = 0.8740; Ratio = $0.8819 \div 0.8740 = 1.0090$. The ratio exceeds 1.00 by 0.90%, confirming an arbitrage opportunity of that magnitude on any notional traded.

4. Triangular Arbitrage

Triangular arbitrage exploits an inconsistency among three exchange rate quotes. If the product of the three effective multipliers — one for each leg of a USD $\rightarrow$ Currency B $\rightarrow$ Currency C $\rightarrow$ USD round trip — differs from 1.0000, a risk-free profit exists. The arbitrageur executes all three legs simultaneously, earning the gap without bearing any market risk.

The no-arbitrage condition requires: $f_{\text{USD} \rightarrow \text{B}} \times f_{\text{B} \rightarrow \text{C}} \times f_{\text{C} \rightarrow \text{USD}} = 1.0000$, where each f_{leg} is the effective multiplier for that leg (which may be the raw quote or its reciprocal, depending on the direction of conversion).

WORKED EXAMPLE 3

Triangular Arbitrage: USD $\rightarrow$ GBP $\rightarrow$ EUR $\rightarrow$ USD (Notional \$1,000,000)

Step	Action & Calculation	Running Balance
Start	Initial capital	\$1,000,000 USD
① USD $\rightarrow$ GBP	$\$1,000,000 \div 1.2700$ (GBP/USD spot)	787,402 GBP
② GBP $\rightarrow$ EUR	$787,402 \div 0.8740$ (market EUR/GBP — underpriced!)	900,918 EUR
③ EUR $\rightarrow$ USD	$900,918 \times 1.1200$ (EUR/USD spot)	\$1,009,028 USD
Net profit	$\$1,009,028 - \$1,000,000$	+\$9,028 risk-free
Product check	$(1 \div 1.12) \times (1 \div 0.874) \times 1.27$	$= 1.0090 \neq 1.0000$ ✓

The 0.90% gap ($= 1.0090 - 1.0000$) equals exactly the profit rate: $\$9,028 / \$1,000,000 = 0.903\%$. The trade is self-correcting: buying EUR bids up EUR/USD, and selling EUR/GBP pushes it toward 0.8819 until the gap closes. In electronic markets this occurs within milliseconds.

IN-CLASS ACTIVITY · Live Arbitrage Simulator

During class, the interactive triangular-arbitrage simulator allowed students to adjust three exchange rate quotes manually and observe the real-time P&L; on a \$1,000,000 notional. The "Inject Arb" button introduced a deliberate mispricing to illustrate how the product $f_{\text{USD}} \times f_{\text{EUR}} \times f_{\text{GBP}}$ deviates from 1.0000 and generates a profit. The key insight is that any price discovery mechanism — including this manual manipulation — immediately creates an incentive for arbitrageurs to trade the gap away.

5. FX Market Manipulation: The WM/Reuters Cartel

Between 2007 and 2013, traders at five major banks — Barclays, JPMorgan, Citigroup, UBS, and RBS — coordinated to manipulate the WM/Reuters 4 p.m. London Fix, the benchmark used to value trillions of dollars in international transactions daily.

The Mechanism

- Accumulate: Traders collected client orders in the hours before the 4 p.m. fix window.
- Share Intel: Net positions were disclosed across competing banks via Bloomberg chat rooms.
- Flood the Fix: At 15:59:30 UTC, coordinated buy or sell orders were placed to move the benchmark in the direction profitable for the banks' net positions.
- Pocket the Spread: Clients were executed at the rigged rate; the difference accrued to the banks.

Fines and Consequences

Bank	Fine (USD)	Regulators
Barclays	\$2.4 billion	DOJ, FCA, NY DFS
Citigroup	\$1.3 billion	DOJ, CFTC
JPMorgan	\$1.0 billion	DOJ, Fed, OCC
UBS	\$800 million	DOJ, CFTC, FINMA
RBS	\$669 million	DOJ, CFTC
Total	≈ \$10 billion	—

Subsequent reforms included widening the fix window to five minutes, mandatory electronic audit trails, prohibitions on cross-bank coordination, and adoption of the Global FX Code of Conduct (2017). The case illustrates that even the deepest, most liquid market in the world (\$6.6 trillion daily turnover in 2022) is vulnerable to manipulation when information is concentrated and oversight is absent. Importantly, arbitrage is the corrective mechanism: any deviation from consistent pricing creates a risk-free profit, attracting trades that restore equilibrium.